GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
ECE 3040A: Microelectronic Circuits
Spring Semester 2026, Test #2
Monday, April 6, 2026

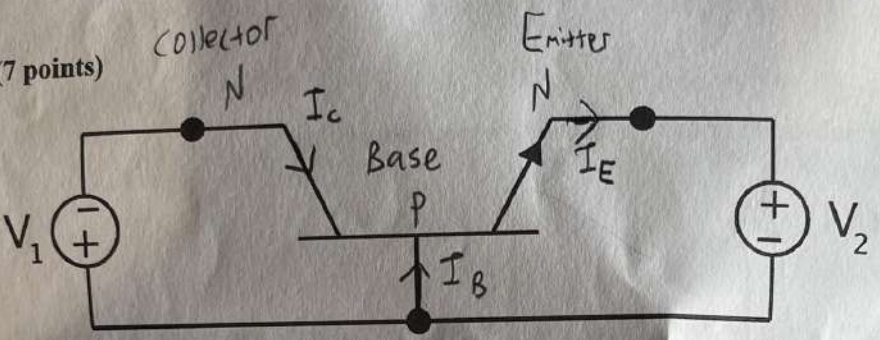
E-B and B-C

~~48~~

50

Name: Henry Garcia

1. **(7 points)**



(a) (1 point) What type of BJT is the transistor shown above? (NPN or PNP)

NPN. ✓

(b) (1 point) Identify emitter, base and collector terminals and clearly show on the figure above. labeled above! ✓

(c) (4 points) Determine the mode of operation under following conditions:

V_1	V_2	Mode of Operation
F Positive	(F) Negative	Saturated ✓
R Negative	(R) Positive	cut-off ✓
F Positive	(R) Positive	Active ✗
R Negative	(F) Negative	inverted ✗

(d) (1 point) Clearly show the **direction of current flow** at the emitter, base and collector terminals on the figure above when the transistor is operated in the **forward active mode**.

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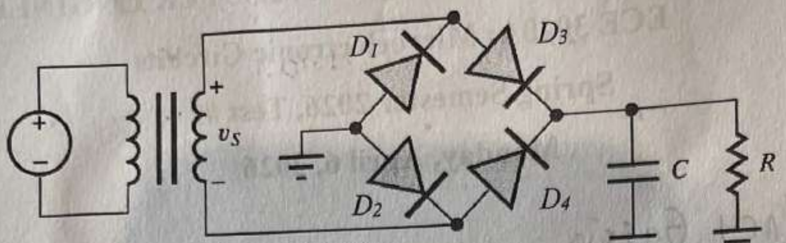
✓

2. **(3.5 points)** In the rectifier circuit given below,



$v_S = V_P \sin \omega t$

(a) (1 points) Which diode or diodes conduct in **positive** cycles?

D_2 and D_3

(b) (1 points) What is the output DC voltage if we assume the turn-on voltage for each diode is 0.7 V?

$V_P - 2V_{on} \Rightarrow V_P - 1.4V$

(c) (3 points) How can you reduce the ripple voltage of this rectifier circuit? Please clearly mark Increase or Decrease for each parameter below.

4

i) Increase or Decrease **C**

ii) Increase or Decrease **ω**

iii) Increase or Decrease **R**

×

3. **(3 points)** Which terminal is the most heavily doped in a BJT and why?

The emitter, because the goal is to ensure that the majority of carriers make it to the collector to maximize current gain.

2

1

4. **(3 points)** Which terminal is the most heavily doped in a MOSFET and why?

In a MOSFET, the source is heavily doped because that is the "metal" and dictates which way the bands will move, determining the mode the MOSFET will operate under.

X

5. **(3 points)** Consider two identical BJTs (BJT1 and BJT2) except that BJT2 has a longer base region. When operated in forward active mode under identical conditions, which one leads to a lower base current? Explain your answer.

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BJT 1 leads to a lower base current, because it's base region is shorter, leading to less recombination and less base current. If the base were longer, more recombination and therefore more base current would occur.

6. **(3 points)** You want to build an PMOS. What type of doped substrate do you need? Between the source and drain in this device, which terminal is biased to have a higher potential in a circuit?

P-MOS,..

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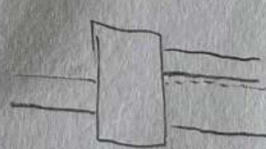
You want an n-type substrate, and want the source to have a higher potential, which drops the metals fermi level, which causes the bands of the semiconductor to bend and go into inversion.



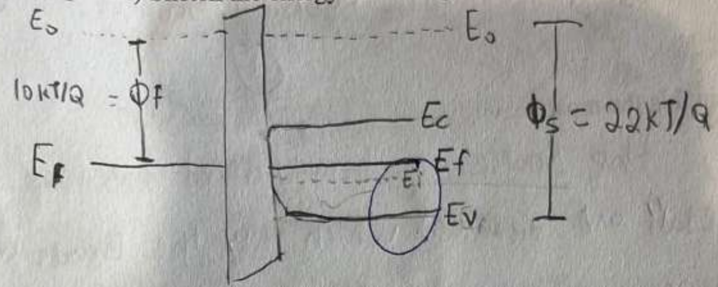
7. **(18 points)** Consider an ideal (flat-band condition) MOS structure for the following conditions.

(a) $\boldsymbol{\phi_F = 10kT/q}$, $\boldsymbol{\phi_S = 22kT/q}$

i) (1 point) Indicate the semiconductor doping type: n-type ✗

ii) (1 point) Biasing regime: depletion ✗

iii) (4 points) Sketch the energy band diagram for the MOS structure



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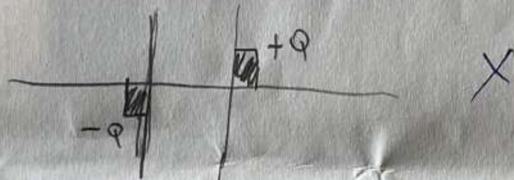
iv) (3 points) Sketch charge diagram for the MOS structure



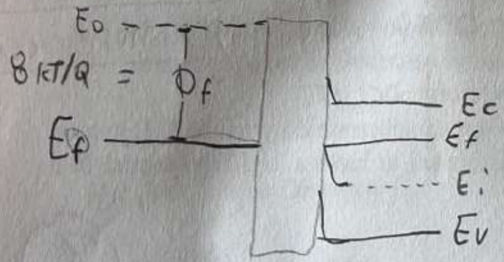
✗

(b) $\boldsymbol{\phi_F = 8kT/q}$, $\boldsymbol{\phi_S = -16kT/q}$

i) (1 point) Indicate the semiconductor doping type: p-type ✓ 1

ii) (1 point) Biasing regime: inversion ✗

iii) (4 points) Sketch the energy band diagram for the MOS structure



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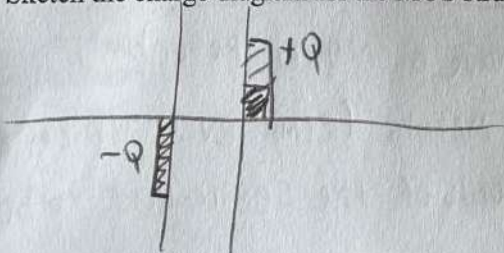
iv) (3 points) Sketch the charge diagram for the MOS structure



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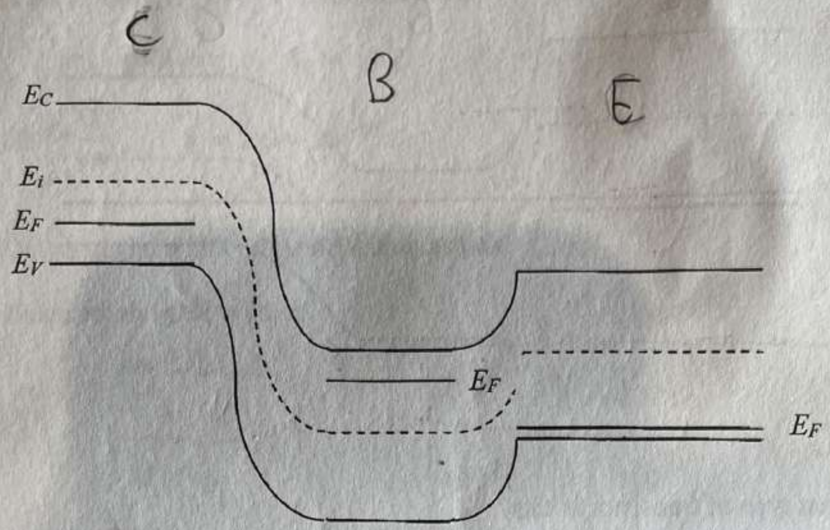
8. **(4 points)** Now, consider the following energy band diagram for a transistor



(a) (1 point) What type of transistor is this?

PNP ✓

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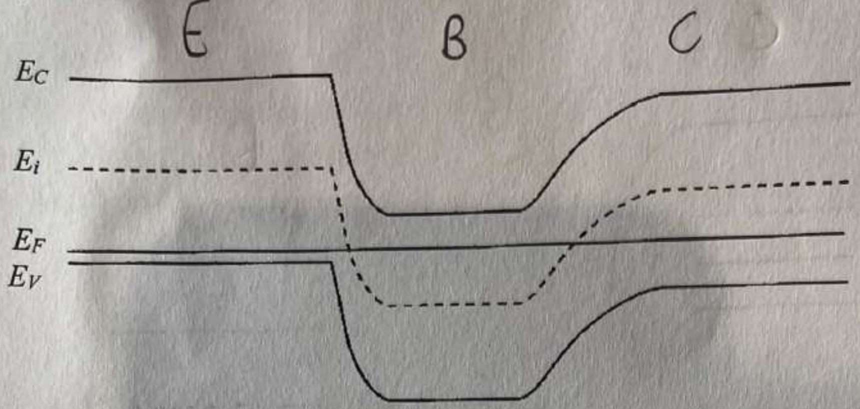
(b) *(1 point)* Clearly mark three transistor terminals on the plot

Marked above. ✓

(c) (2 points) What is the mode of operation for the transistor, whose energy band diagram in given in the above plot?

Active ✓

9. **(4 points)** Consider the following energy band diagram for a transistor



(a) (1 point) What type of transistor is this?

PNP ✓

(b) (1 point) Clearly mark three transistor terminals on the plot

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Marked above! ✓

(c) (2 points) What is the mode of operation for the transistor, whose energy band diagram in given in the above plot?

Inverted ✗

Problem 1 (10 points). A MOS capacitor has its silicon semiconductor region doped with boron at a concentration of $1 \times 10^{16}\ cm^{-3}$. Given also that:

$\phi_s = -0.5\ V$, $\quad x_o = 0.5\ \mu m$, $\quad K_S = 11.7$, $\quad K_O = 3.9$, $\quad T = 300\ K$,

$\epsilon_0 = 8.85 \times 10^{-14} \frac{F}{cm}$, $\quad q = 1.6 \times 10^{-19}\ C$

(a) (5 points) Calculate the width of the depletion region.

(b) (5 points) Calculate the gate voltage.

$$a)\quad W = \sqrt{\frac{2k_S\varepsilon_0}{q \cdot N_A}\phi_S} = \sqrt{\frac{2(11.7)(8.85\cdot 10^{-14})}{(1.6\cdot 10^{-19})\cdot(1\cdot 10^{16})}\cdot(-0.5)}$$

$$= \sqrt{\frac{2.0709\cdot 10^{-12}}{0.0016}\cdot(0.5)} = \underline{2.543\,\mu M} \quad \times \quad 3$$

$$b)\quad V_g = \phi_S + \frac{k_S}{k_O}x_o\sqrt{\frac{2qN_A}{k_S\varepsilon_0}\phi_S}$$

$$= \cancel{+}0.5 + \frac{11.7}{3.9}\cdot 0.5\,\mu m\sqrt{\frac{2\cdot 1.6\cdot 10^{-9}\cdot 1\cdot 10^{16}}{11.7\cdot 8.85\cdot 10^{-14}}\cdot 0.5}$$

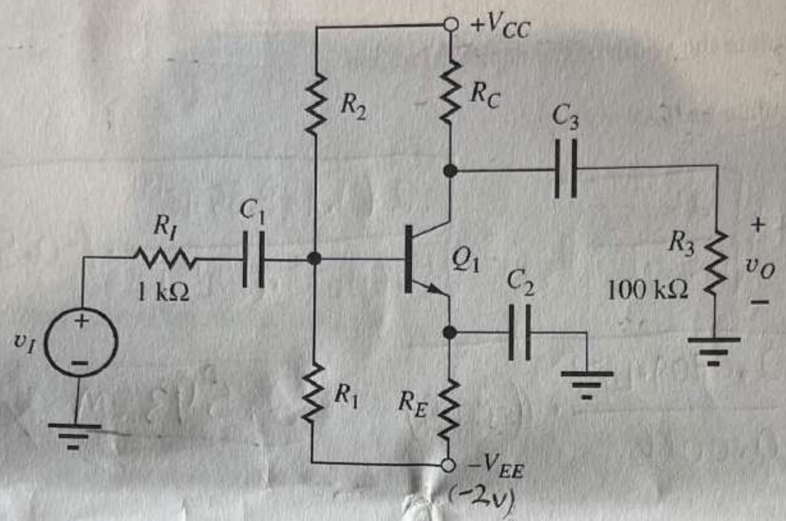
$$= 0.5 + (3\cdot 0.5\,\mu M)\cdot\sqrt{\frac{32{,}000{,}000}{1.035\cdot 10^{-12}}\cdot 0.5}$$

$$= \underline{5.8}\ \text{unit?} \quad 3$$

Problem 2 (40 points): For the circuit shown on the right, assume the transistor is operated in the active mode with the following parameters:

$\beta_F = 100$, $V_{ON} = 0.75\ V$, $V_A = 110\ V$, $R_1 = 36\ k\Omega$, $R_2 = 72\ k\Omega$,

$R_C = 110\ k\Omega$, $R_E = 120\ k\Omega$, $V_{CC} = 12\ V$, $V_{EE} = 2\ V$



(a) (5 points) Draw the DC equivalent circuit.

(b) (5 points) First find the Thevenin equivalent of the base bias circuit and simplify the equivalent DC circuit.

(c) (10 points) Find the Q-point (I_C, V_{CE}) of the BJT.

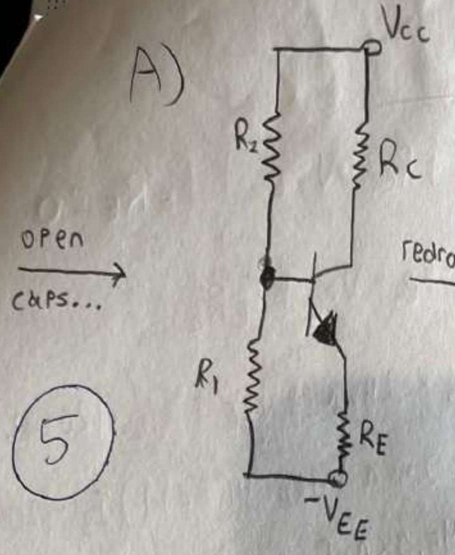
(d) (5 points) Draw the AC equivalent circuit and simplify the circuit by calculating its Thevenin equivalent.

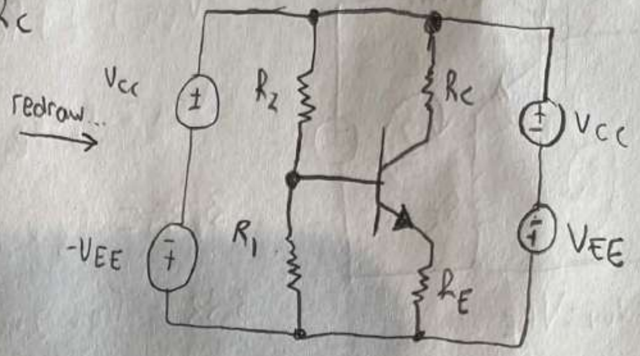
(e) (5 points) Calculate the small signal equivalent parameters and replace the transistor with a small signal equivalent circuit.

(f) (10 points) Calculate the signal source voltage gain v_O/v_I

A)

For Part A, all capacitors at DC act like an open circuit/wire.





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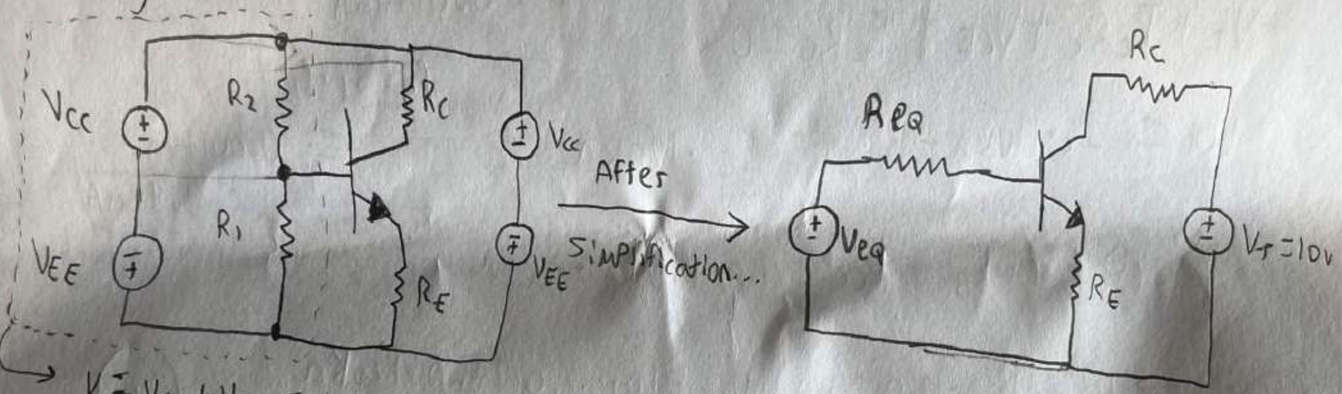
B)



$V_T = V_{CC} + V_{EE} = 12 - 2 = 10V$

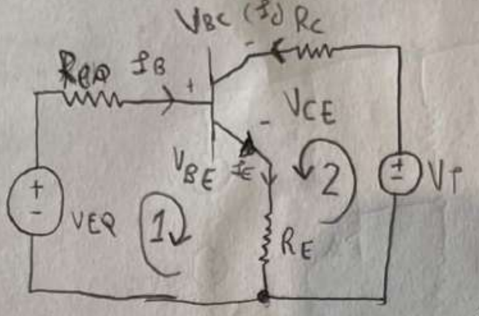
$V_{th} = V_{eq} = 10V \cdot \frac{R_1}{R_2 + R_1} = 10V \cdot \frac{36k\Omega}{36k\Omega + 72k\Omega} = 3.33V$ ✗

3

$R_{th} = R_2 || R_1 = \frac{36k\Omega \cdot 72k\Omega}{36k\Omega + 72k\Omega} = 24{,}000\Omega$ ✓

$(24k\Omega)$

c) Need to find (I_C, V_{CE})



Note: $V_{ON} = V_{BE}$

loop 1:

$-V_{eq} + I_B R_{eq} + V_{BE} + I_E R_E = 0$

$V_{eq} = I_B \cdot R_{eq} + V_{BE} + I_E R_E$

$3.33 = I_B \cdot 24k\Omega + V_{BE} + I_E \cdot 120k\Omega$

$3.33 = I_B \cdot 24k\Omega + V_{BE} + ((\beta_f + 1) I_B) \cdot 120k\Omega$

$\rightarrow 3.33 = 145.2k\Omega \cdot I_B + 0.75V$

$\rightarrow I_B = 0.0178mA$

$I_C = \beta_f \cdot I_b$

$= 0.00178A$

$= \boxed{1.78mA}$

loop 2:

$-V_T + I_C R_C + V_{CE} + I_E R_E = 0$

$10 = I_C \cdot 110k\Omega + V_{CE} + I_E \cdot 120k\Omega$

$\rightarrow 10 = \beta_f \cdot I_B + V_{CE} + (\beta_f + 1) I_B \cdot 120k\Omega$

$\rightarrow 10 = 100 \cdot 0.0178mA + V_{CE} + 101 \cdot 0.0178mA \cdot 120k\Omega$

$\rightarrow 10 = 0.00178A + V_{CE} + 0.0017978_{mA} \cdot 120k\Omega$

$V_{CE} = V_{BE} - V_{BC}$

$V_{CE} = -205.736V$

Q-Point: $(1.78mA, -205.736V)$

X

4